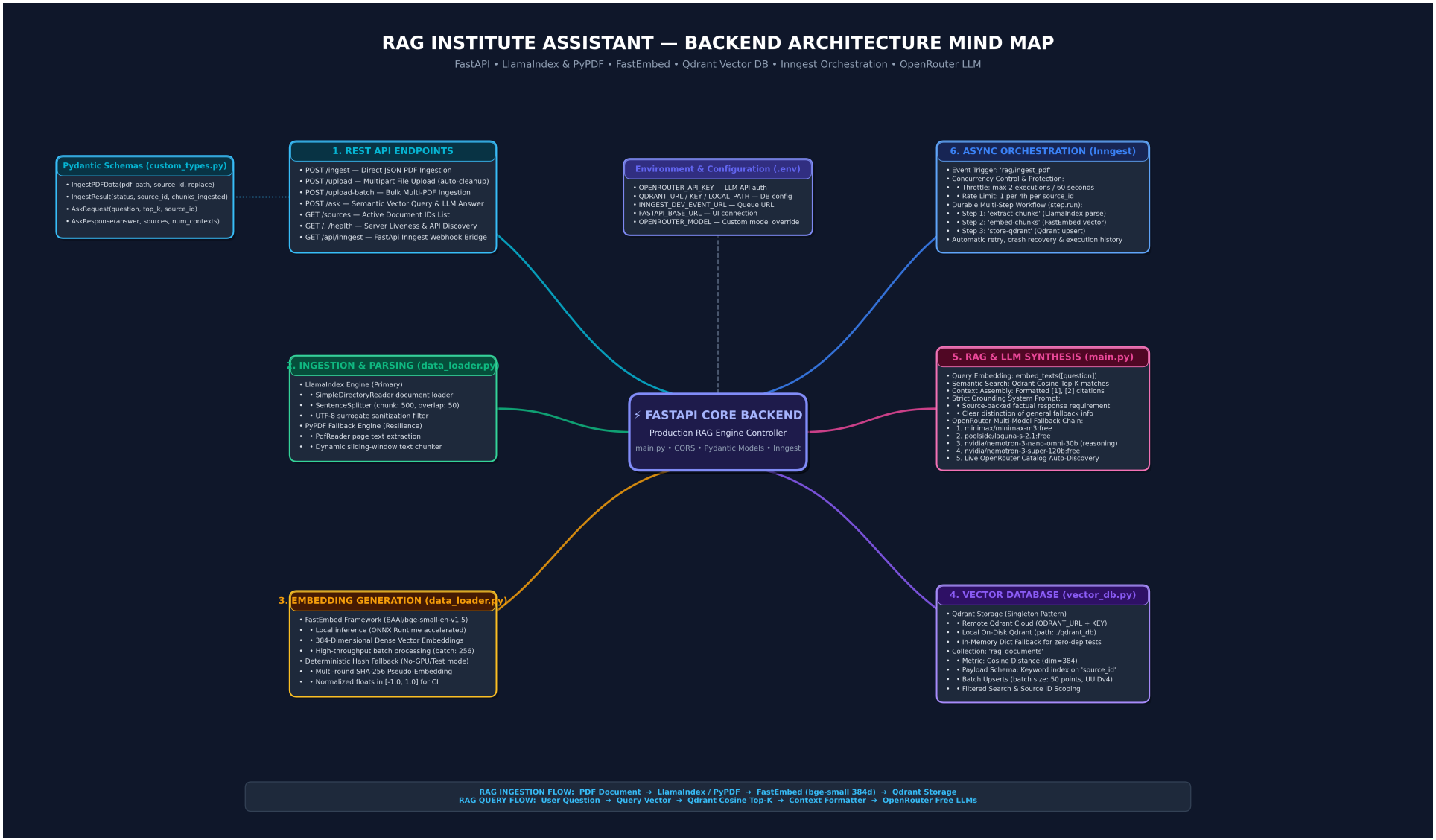
RAG Institute Assistant — Backend Architecture Mind Map

Visual hierarchy of FastAPI services, document parsing, embeddings, vector indexing, Inngest orchestration, and LLM synthesis.



Backend Architectural Components & Data Flow

Comprehensive breakdown of backend modules, endpoints, resilient fallbacks, and execution pipelines.

Component / Layer	Implementation File	Primary Technologies	Key Responsibilities & Fallbacks
1. API Controller	`main.py`	FastAPI, Uvicorn, Pydantic, CORS	Exposes REST endpoints (`/ingest`, `/upload`, `/upload-batch`, `/ask`, `/sources`, `/health`). Handles input validation and orchestrates pipelines.
2. Parsing & Chunking	`data_loader.py`	LlamaIndex (SentenceSplitter) PyPDF (Fallback)	Extracts text from PDF documents. Chunks text into 500-token windows with 50-token overlap. Sanitizes UTF-8 surrogate characters. Falls back to PyPDF sliding window.
3. Vector Embeddings	`data_loader.py`	FastEmbed (bge-small-en-v1.5) Deterministic SHA-256 (Fallback)	Generates 384-dimensional dense vectors using local ONNX inference. When offline or in CI test mode, computes deterministic SHA-256 float vectors without ML models.
4. Vector Storage	`vector_db.py`	Qdrant Client (Cloud / Disk / In-Memory)	Singleton storage wrapper. Manages `rag_documents` collection with Cosine similarity. Supports batch upserts (50/batch), keyword indexing on `source_id`, and metadata filtering.
5. Async Orchestration	`main.py`	Inngest SDK (Event-driven)	Orchestrates background document ingestion with durability. Configured with 2-per-minute throttle and 1-per-4h rate limit per source. 3-step pipeline: chunking ➔ embedding ➔ storage.
6. LLM Synthesis	`main.py`	OpenRouter API, OpenAI Client	Constructs context blocks with [1], [2] citations. Executes resilient fallback chain across free models (Gemma-4 31B, Nemotron Nano 9B, Gemma-4 26B, GPT-OSS 20B).

End-to-End Data Lifecycle

• **Ingestion Pipeline:** `PDF Upload ➔ Temp Storage ➔ LlamaIndex/PyPDF Token Chunker ➔ FastEmbed 384d Generator ➔ Qdrant Batch Upsert ➔ Temp Cleanup`
• **Query Pipeline:** `User Question ➔ Query FastEmbed Vector ➔ Qdrant Cosine Similarity Search ➔ Context Assembly with Citations ➔ OpenRouter LLM Call ➔ Grounded Answer`